Jupyter Notebook Execution Report

Name: Your Name

Date: January 06, 2026

Cell 1: ■ Markdown

Assignment 6

Cell 2: ■ Code

```
import torch
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

Cell 3: ■ Markdown

PART 1: Data Generation (The Exclusion Zone)

Cell 4: ■ Code

```
GROUP_NUMBER = 8
```

```
# Manual seed
torch.manual_seed(GROUP_NUMBER)
np.random.seed(GROUP_NUMBER)
```

```
def generate_data(n_samples=3000):
# Generate random coordinates between -5 and 5 km
X = (torch.rand(n_samples, 2) * 10) - 5
```

```
# Calculate distance from center (radius)
radius = torch.sqrt(X[:, 0]**2 + X[:, 1]**2)
```

```
# Assign Labels: 1 if inside the Dead Zone, 0 otherwise
y = ((radius &gt; 2) & (radius &lt; 4)).float().view(-1, 1)
```

```
# Add 5% Noise (Malfunctioning Drones)
n_noise = int(0.05 * n_samples)
noise_indices = torch.randperm(n_samples)[:n_noise]
y[noise_indices] = 1 - y[noise_indices] # Flip labels
```

```
return X, y
```

```
# Generate the dataset
X_full, y_full = generate_data(3000)
```

```
print(f"Data Generated: {X_full.shape} samples.")
print(f"Target Generated: {y_full.shape} labels.")
```

Output:

```
Data Generated: torch.Size([3000, 2]) samples.
Target Generated: torch.Size([3000, 1]) labels.
```

Cell 5: ■ Markdown

PART 2: Splitting the Data & Parameter Initialization

Cell 6: ■ Code

```
# Split Data (70% Train, 30% Val)
indices = torch.randperm(X_full.size(0))
train_size = int(0.7 * X_full.size(0))
```

```
X_train, y_train = X_full[indices[:train_size]], y_full[indices[:train_size]]
X_val, y_val = X_full[indices[train_size:]], y_full[indices[train_size:]]
```

Cell 7: ■ Code

```
# Network Architecture
def init_weights(shape):
return (torch.randn(shape) * torch.sqrt(torch.tensor(2.0 /
shape[0]))).requires_grad_(True)
```

```
def init_bias(shape):
return torch.zeros(shape, requires_grad=True)
```

```
# Layer 1
W1 = init_weights((2, 16))
b1 = init_bias((1, 16))
```

```
# Layer 2
W2 = init_weights((16, 16))
b2 = init_bias((1, 16))
```

```
# Layer 3
W3 = init_weights((16, 16))
b3 = init_bias((1, 16))
```

```
# Output Layer
W4 = init_weights((16, 1))
b4 = init_bias((1, 1))
```

```
parameters = [W1, b1, W2, b2, W3, b3, W4, b4]
```

Cell 8: ■ Markdown

Here I have used He Initialization (std = sqrt(2/fan_in))

- The weights are plain tensors with requires_grad=true means they will accumulate gradients and can be updated manually

Cell 9: ■ Markdown

PART 3: Activation & Loss Functions

Cell 10: ■ Code

```
def relu(z):
return z.clamp(min=0)
```

```
def sigmoid(z):
return 1 / (1 + torch.exp(-z))
```

```
def forward_pass(X):
z1 = torch.matmul(X, W1) + b1
a1 = relu(z1)
```

```
z2 = torch.matmul(a1, W2) + b2
a2 = relu(z2)
```

```
z3 = torch.matmul(a2, W3) + b3
a3 = relu(z3)
```

```
z4 = torch.matmul(a3, W4) + b4
y_hat = sigmoid(z4)
return y_hat
```

Cell 11: ■ Code

```
def binary_cross_entropy(y_pred, y_true):
# Clamp to avoid log(0) which leads to NaN
y_pred = torch.clamp(y_pred, 1e-7, 1 - 1e-7)
loss = -torch.mean(y_true * torch.log(y_pred) + (1 - y_true) * torch.log(1 -
y_pred))
return loss
```

Cell 12: ■ Markdown

PART 4: Training Loop

Cell 13: ■ Code

```
learning_rate = 0.1
epochs = 2000
```

```
train_losses = []
val_losses = []
train_accs = []
val_accs = []
```

```
for epoch in range(epochs):
```

```
# Forward Pass
y_pred_train = forward_pass(X_train)
train_loss = binary_cross_entropy(y_pred_train, y_train)
```

```
y_pred_val = forward_pass(X_val)
```

```
val_loss = binary_cross_entropy(y_pred_val, y_val)
```

```
# Accuracy
train_acc = ((y_pred_train &gt; 0.5) == y_train).float().mean()
val_acc = ((y_pred_val &gt; 0.5) == y_val).float().mean()
```

```
# Backpropagation
if any(p.grad is not None for p in parameters):
for p in parameters:
p.grad.zero_()
```

```
train_loss.backward()
```

```
# Manual Gradient Descent
with torch.no_grad():
for p in parameters:
p -= learning_rate * p.grad
```

```
train_losses.append(train_loss.item())
val_losses.append(val_loss.item())
train_accs.append(train_acc.item())
val_accs.append(val_acc.item())
```

Cell 14: ■ Markdown

PART 5: GDC HUD (Visualization Dashboard)

Cell 15: ■ Code

```
fig, ax = plt.subplots(1, 3, figsize=(18, 5))
```

```
# 1. System Error Trajectory (Loss)
ax[0].plot(train_losses, label='Train Loss')
ax[0].plot(val_losses, label='Validation Loss')
ax[0].set_title("System Error Trajectory")
ax[0].set_xlabel("Epoch")
ax[0].set_ylabel("Loss")
ax[0].legend()
```

```
# 2. Integrity Check (Accuracy)
ax[1].plot(train_accs, label='Train Accuracy')
ax[1].plot(val_accs, label='Validation Accuracy')
ax[1].set_title("Integrity Check (Accuracy)")
ax[1].set_xlabel("Epoch")
ax[1].set_ylabel("Accuracy")
ax[1].legend()
```

```
# 3. Geospatial Threat Map (Decision Boundary)
h = 0.1
x_min, x_max = -5.5, 5.5
y_min, y_max = -5.5, 5.5
```

```
xx, yy = np.meshgrid(
np.arange(x_min, x_max, h),
np.arange(y_min, y_max, h)
)
```

```
grid = torch.tensor(
np.c_[xx.ravel(), yy.ravel()],
dtype=torch.float32
)
```

```
with torch.no_grad():
probs = forward_pass(grid).reshape(xx.shape)
```

```
ax[2].contourf(xx, yy, probs, levels=[0, 0.5, 1],
colors=['green', 'red'], alpha=0.3)
```

```
ax[2].scatter(
X_val[:, 0], X_val[:, 1],
c=y_val.flatten(),
cmap='bwr',
s=10,
alpha=0.6
)
```

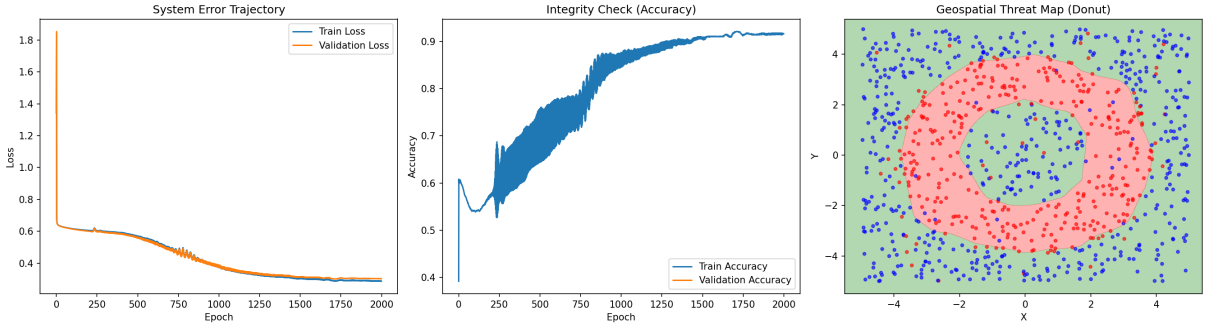
```
ax[2].set_title("Geospatial Threat Map (Donut)")
```

```
ax[2].set_xlabel("X")
ax[2].set_ylabel("Y")
```

```
plt.tight_layout()
plt.show()
```

Output:

```
[STDERR]
&lt;string&gt;:1: UserWarning: FigureCanvasAgg is non-interactive, and thus cannot be shown
```



Cell 16: ■ Markdown

Conclusion

- The decision boundary forms a closed non-linear ring, demonstrating that the MLP successfully learned the implicit radial geometry despite label noise and without explicit distance features